

# Rodin: 3D Asset Creation with a Text/Image/3D-Conditioned Large Generative Model for Creative Frontier

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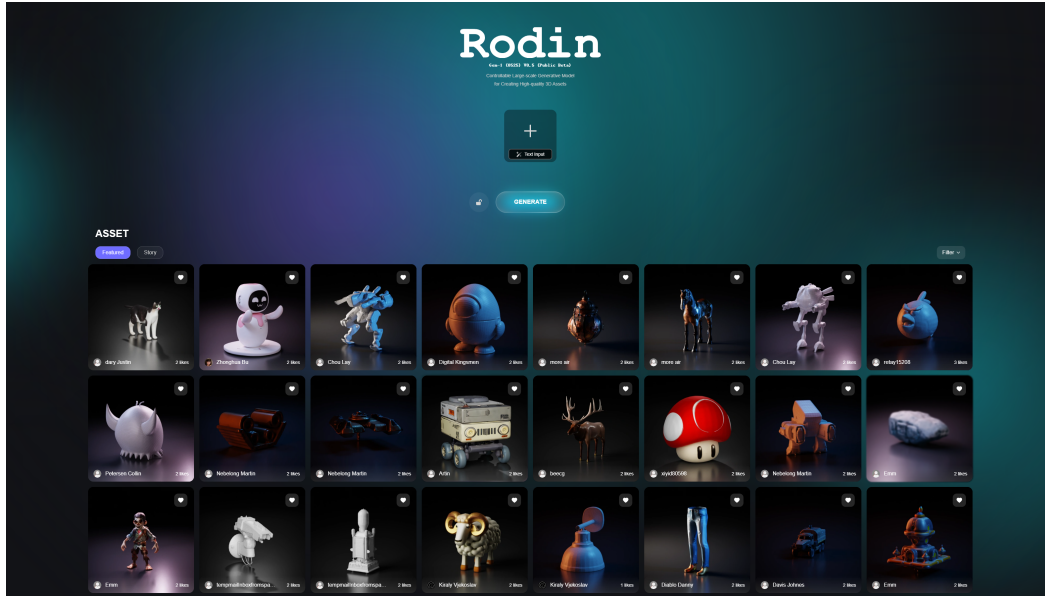


Figure 1: Preview of Rodin.

## ABSTRACT

"Rodin" leverages a Latent Diffusion Transformer and a 3D ConditionNet to facilitate the rapid generation of 3D assets from text, images, and direct inputs. This system streamlines the creation process, producing production-ready models optimized for real-time engines. This "Real-Time Live!" highlights Rodin's innovative framework, its seamless integration into professional workflows, and its transformative impact on the 3D content creation industry.

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## 1 INTRODUCTION

"Rodin" revolutionizes 3D asset creation by leveraging a text/image/3D-conditioned large generative model to produce high-quality, real-time 3D assets. This system combines a Latent Diffusion Transformer (DiT) architecture with a 3D ConditionNet, enabling intuitive multi-modal input and automated production of ready-to-use models. Rodin's key contributions include:

- Versatile Input Acceptance: It processes text, images, and 3D sketches, enhancing accessibility and creative freedom.

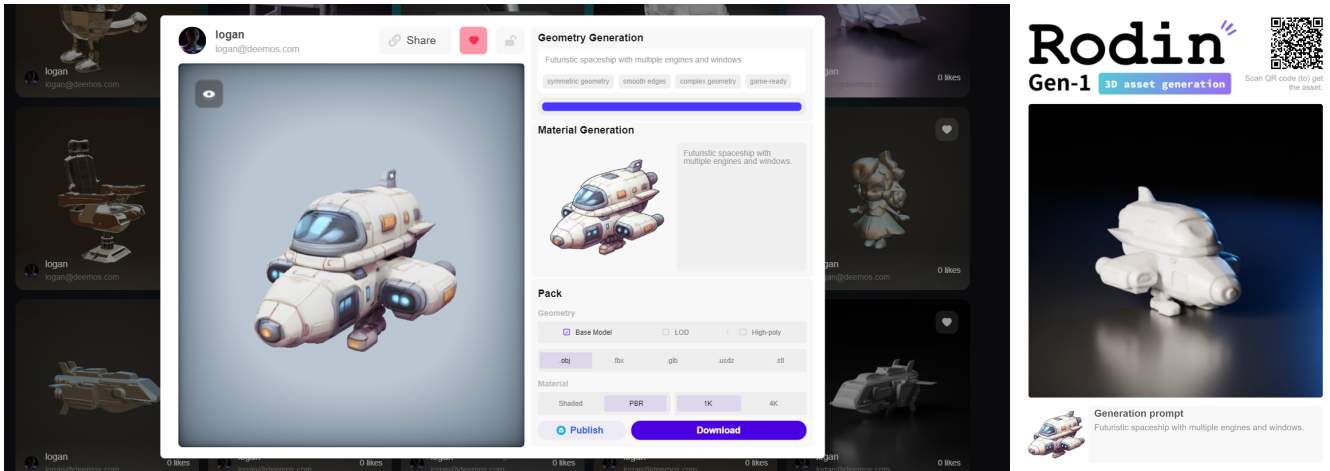


Figure 2: User interface and sharing poster.

- Efficient, High-Quality Output: Generates quad-faced models with PBR textures that meet professional production standards.
- Enhanced Creative Workflow: Automates complex modeling tasks, allowing creators to focus on design and innovation.

## 2 TECHNICAL FRAMEWORK

Rodin’s core is defined by its integration of the Latent Diffusion Transformer (DiT) with a sophisticated 3D ConditionNet, which collectively facilitate a seamless and dynamic 3D asset creation process. The DiT architecture specializes in synthesizing detailed 3D models efficiently, capturing nuances typically missed by conventional methods. This efficiency is crucial for maintaining real-time performance standards.

The system’s 3D ConditionNet enhances user control, allowing precise manipulation of the generated assets’ dimensions and properties based on real-time input variations. This control is pivotal for artists needing to tweak designs on the fly during live sessions or under tight deadlines. Furthermore, Rodin’s ability to handle multiple input types simultaneously revolutionizes traditional 3D modeling workflows, offering a platform that is not only powerful in terms of computational capabilities but also exceptionally adaptive to creative needs.

Together, these technologies ensure that Rodin not only matches but often surpasses the quality and functionality of existing 3D asset creation tools, providing a robust solution that is both technologically advanced and intuitively aligned with user expectations. This dual focus on innovation and usability underscores Rodin’s potential to redefine industry standards for 3D content generation.

## 3 APPLICATION AND INTEGRATION

Rodin excels in transforming the interaction between users and 3D asset creation. It accommodates inputs from text descriptions, images, and direct 3D manipulations, making the process accessible to a diverse user base from seasoned professionals to hobbyists. This inclusivity broadens the scope of creative expression and significantly reduces the learning curve associated with 3D modeling.

The true power of Rodin lies in its seamless integration with existing production pipelines. Assets generated are immediately compatible with major real-time engines such as Unity and Unreal Engine, facilitating a smooth transition from creation to implementation. This compatibility ensures that assets are not only detailed and aesthetically pleasing but also optimized for performance, ready for immediate deployment in various digital environments including gaming, virtual reality, and animation.

Moreover, Rodin enhances traditional workflows by automating the most complex aspects of asset creation. This automation allows artists to concentrate more on the creative aspects of their projects, thereby increasing productivity and fostering innovation. By minimizing technical barriers and maximizing creative output, Rodin empowers creators to bring their visions to life with unprecedented speed and flexibility.

## 4 IMPACT AND FUTURE POTENTIAL

Rodin not only advances the current state of 3D asset creation but also sets the stage for future innovations in the industry. By automating intricate modeling tasks, Rodin frees artists to focus on creativity and design, thus accelerating project timelines and enhancing the quality of digital experiences. Its impact is particularly significant in industries where speed and efficiency are paramount, such as video game development and film production.

Looking ahead, the potential applications of Rodin are vast. As the technology matures, we anticipate enhancements that will introduce even more sophisticated AI-driven interactions, broader asset types, and improved integration with augmented and virtual reality platforms. These developments promise to further democratize 3D content creation, making it more accessible to non-experts and fostering a new era of collaborative and interactive media creation.

Rodin’s ongoing evolution will continue to influence and inspire both the technology and creative sectors, driving forward the boundaries of what is possible in real-time digital content creation. As we showcase Rodin at SIGGRAPH 2024, we invite the community to explore its capabilities and consider its role in shaping the future of our digital world.